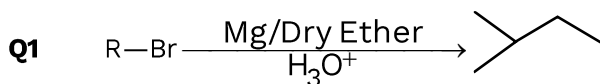
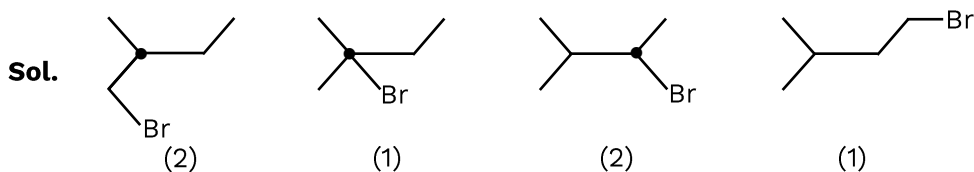


CHEMISTRY



Count the number of possible reactant for given compound?

Ans. (6)



Q2 Arrange the following in increasing order of stability

- (I) $[Co(NH_3)_4]^{2+}$ (II) $[Co(NH_3)_6]^{3+}$ (III) $[Co(NH_3)_6]^{2+}$ (IV) $[Co(en)_3]^{3+}$
 (1) $IV > II > III > I$ (2) $IV > III > II > I$ (3) $I > IV > II > III$ (4) $I > II > III > IV$

Ans. (1)

Sol. → Octahedral have more splitting CFSE than tetrahedral
 → Strong field ligand
 → More CFSE
 → More the charge on metal more splitting CFSE
 So overall: $IV > II > III > I$.

Q3 Density of 3M NaOH is 1.25 g/ml. Molality of solution is:

- (1) 2.65 (2) 2.5 (3) 2.8 (4) 3

Ans. (1)

Sol. $D_{\text{solution}} = 1.25 \text{ g/ml}$
 $3M \rightarrow 3 \text{ mole solute in } \underbrace{1 \text{ litre solution}}_{1000 \text{ ml}}$

$$D = \frac{M}{V}$$

$$\text{Solution (M)} = D \times V = 1.25 \times 1000 = 1250 \text{ g}$$

$$\text{Solute Mass} = n \times \text{Molar mass}$$

$$= 3 \times 40 = 120 \text{ g}$$

$$NaOH \ 23 + 16 + 1 \Rightarrow 40 \Rightarrow \text{Molar mass}$$

$$M_{\text{solution}} = M_{\text{solute}} + M_{\text{solvent}}$$

$$= 1250 - 120 \Rightarrow 1130 \text{ g}$$

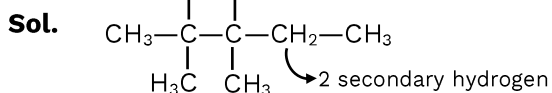
$$\text{Molality}(m) = \frac{n}{M_{\text{solvent}}(\text{kg})}$$

$$= \frac{3 \times 1000}{1130} \Rightarrow 2.65$$

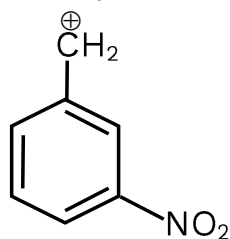
Q4 Which of the following has “two secondary hydrogens”

- (1) 2,2,3,3-Dimethyl pentane
- (2) 2,2,4,4-Dimethyl heptane
- (3) 4-Ethyl-2,2-Dimethyl Hexane
- (4) None of these

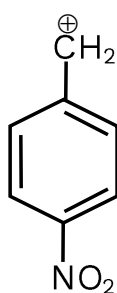
Ans. (1)



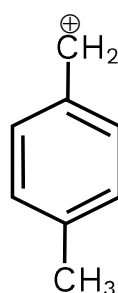
Q5 Stability of carbocation is



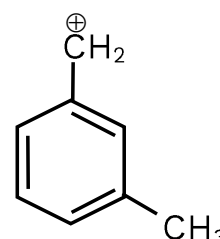
(P)



(Q)



(R)



(S)

(1) $R > S > P > Q$

(2) $R > P > S > P$

(3) $S > Q > R > P$

(4) $R > P > S > Q$

Ans. (1)

Sol. $R > S > P > Q$

Stability of carbocation increase with electron donating group and decrease with withdrawing group.

Q6 Assertion: Element from left of the periodic table are acidic in nature.

Reason: Element oxide from right of the periodic table on hydrolysis gives acid.

- (1) Both A and R are correct and R is the correct explanation of A.
- (2) Both A and R are correct, but R is not the correct explanation of A.
- (3) A is correct but, R is incorrect.
- (4) R is correct but, A is incorrect.

Ans. (4)

Q7 Which of the following does not show disproportionation reaction

- (1) ClO_4^-
- (2) ClO_3^-
- (3) ClO_2^-
- (4) ClO^-

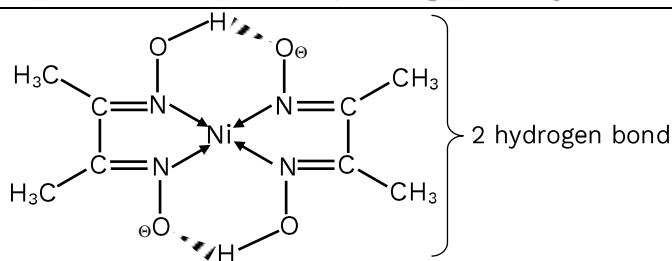
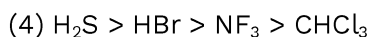
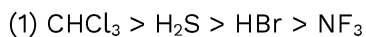
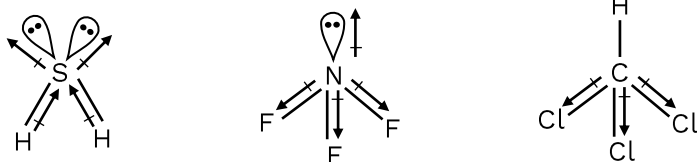
Ans. (1)

Sol. In ClO_4^- Cl is in maximum oxidation state i.e. +7

Q8 Total number of H-atom in complex



Ans. (2)

Sol.**Q9** Order of dipole moment of:**Ans.** (1)**Sol.****Q10** Which of the following group 15 element covalency with minimum A—A bond energy.**Ans.** (4)**Sol.** Bond energy determined by bond strength so Sb—Sb due to high size have least bond energy**Q11** Given below are two statements:**Statement-1:** In corrosion of metal, the metal acts as cathode**Statement-2:** Alkaline medium increases rate of corrosion.

In the light of the above statements, choose the most appropriate answer from the options given below :

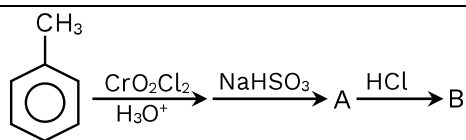
(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

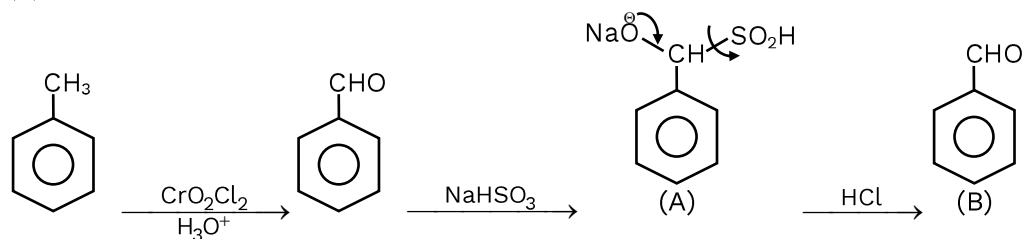
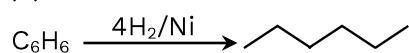
(4) Statement I is incorrect but Statement II is correct

Ans. (1)**Sol.** Both statement (1) and (2) incorrect

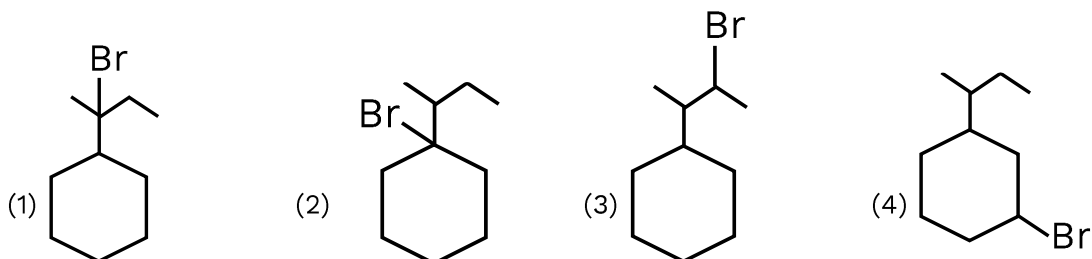
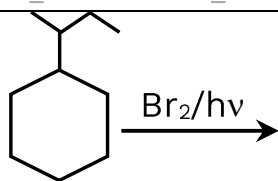
Q12

The product is:

- (A) (A) (B)
- (2) (A) (B)
- (3) (A) (B)
- (4) (A) (B)

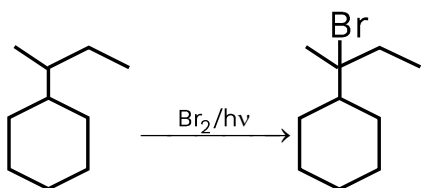
Ans. (2)**Sol.****Q13** In the reaction of C_6H_6 in presence of H_2/Ni , 4 mole of H_2 is consumed, number of possible π -bonds will be:**Ans.** (4)**Sol.**i.e 4 π bond

Q14

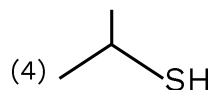
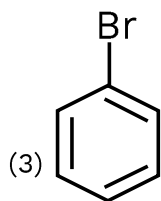
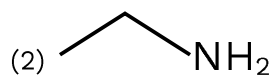


Ans. (1)

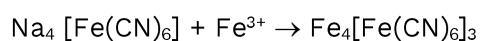
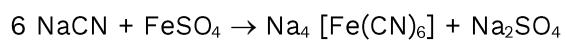
Sol.

Free radical mechanism $3^\circ > 2^\circ > 1^\circ$

Q15 In given compound which one give Prussian blue colour lassaigne's test

(1) NH_2NH_2 

Ans. ()

Sol. $\text{Na} + \text{C} + \text{N} \rightarrow \text{NaCN}$ 

(Prussian blue)

Thus, we need C + N

Q16 200 ml 0.2 M solution of NaOH is mixed with 400 ml of 0.5 M NaOH solution. Molarity of mixture is

(1) 0.4 M

(2) 0.6 M

(3) 4 M

(4) 0.8 M

Ans. (1)

Sol. $M_1V_1 + M_2V_2 = M_3V_3$

$$0.2 \times 200 + 0.5 \times 400 = M_3 \times 600$$

$$0.4 + 0.2 = M \times 6$$

$$M = \frac{2.4}{6} = 0.4 \text{ M}$$

Q17 For a diatomic gas if $\gamma_1 = C_p/C_v$ for rigid molecules and $\gamma_2 = C_p/C_v$ for another diatomic molecule having vibration then

- (1) $\gamma_2 < \gamma_1$ (2) $\gamma_2 > \gamma_1$ (3) $\gamma_2 = \gamma_1$ (4) $\gamma_2 = 2\gamma_1$

Ans. (1)

Sol. (i) $C_v = \frac{3R}{2} + \frac{2R}{2} = \frac{5R}{2}$

$$C_p = \frac{7R}{2}$$

$$\gamma = \frac{C_p}{C_v} = \frac{7}{5} = 1.4$$

(ii) $C_v = \frac{3R}{2} + \frac{2R}{2} + R = \frac{7R}{2}$

$$C_p = \frac{9R}{2}$$

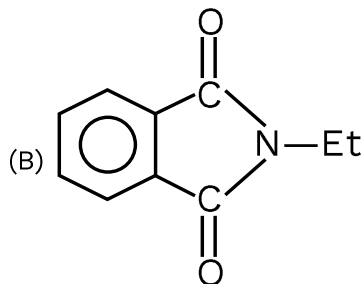
$$\gamma = \frac{9}{7} = 1.28$$

Q18 Which is the correct combination for amine as final product:

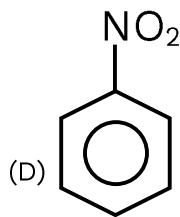
List-I

(Reactant)

(A) $RCONH_2$



(C) $R-CN$



List-II

(Reagent)

(P) aq. NaOH

(Q) $LiAlH_4$

(R) $SnCl_2/HCl$

(S) H_2/Pd

(1) $A \rightarrow R$; $B \rightarrow P$; $C \rightarrow Q$; $D \rightarrow S$

(3) $A \rightarrow P$; $B \rightarrow Q$; $C \rightarrow R$; $D \rightarrow S$

(2) $A \rightarrow Q$; $B \rightarrow P$; $C \rightarrow S$; $D \rightarrow R$

(4) $A \rightarrow S$; $B \rightarrow R$; $C \rightarrow Q$; $D \rightarrow P$

Ans. (2)

Q19 Match the list

List-I

(i) $\left(\frac{dG}{dP}\right)_T$

(ii) $\left(\frac{dG}{dT}\right)_P$

(iii) $\left(\frac{dH}{dT}\right)_P$

(iv) $\left(\frac{dU}{dT}\right)_V$

List-II

(P) V

(Q) C_V

(R) C_P

(S) $-S$

(1) (i) $\rightarrow$ (P), (ii) $\rightarrow$ (S), (iii) $\rightarrow$ (R), (iv) $\rightarrow$ (Q)

(2) (i) $\rightarrow$ (S), (ii) $\rightarrow$ (P), (iii) $\rightarrow$ (R), (iv) $\rightarrow$ (Q)

(3) (i) $\rightarrow$ (R), (ii) $\rightarrow$ (P), (iii) $\rightarrow$ (Q), (iv) $\rightarrow$ (S)

(4) (i) $\rightarrow$ (Q), (ii) $\rightarrow$ (R), (iii) $\rightarrow$ (S), (iv) $\rightarrow$ (P)

Ans. (1)

Q20 The solubility of salt A_3B_4 (s) is:

(1) $\left(\frac{K_{sp}}{6912}\right)^{1/6}$

(2) $\left(\frac{K_{sp}}{6912}\right)^{1/7}$

(3) $\left(\frac{K_{sp}}{6912}\right)^{1/2}$

(4) $\left(\frac{K_{sp}}{6912}\right)^{2/2}$

Ans. (2)

Sol. $A_3B_4 \rightarrow 3A^{4+} + 4B^{3-}$

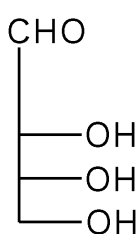
$$K_{sp} = (3S)^3(4S)^4$$

$$= 3^3 4^4 S^{3+4}$$

$$K_{sp} = 27 \times 256 S^7$$

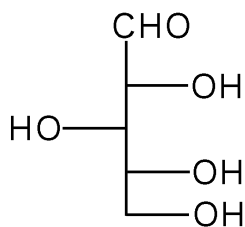
$$S = \left(\frac{K_{sp}}{6912}\right)^{1/7}$$

Q21 Identify number of structure which can be correlated to D-glyceraldehyde.



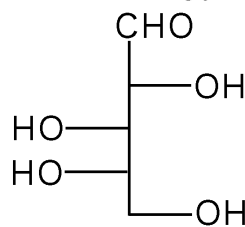
(a)

(1) 2



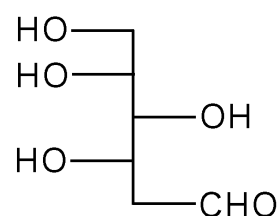
(b)

(2) 1



(c)

(3) 4



(d)

(4) 3

Ans. (2)

Q22 Calculate the radius of first excited state of He^+ ion

(in Å)

(1) 0.529 Å

(2) 1.058 Å

(3) 0.236 Å

(4) 2.116 Å

Ans. (2)

Sol. $r_n = 0.529 \frac{n^2}{Z}$

For He^+ $Z = 2$,

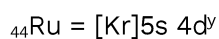
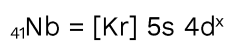
$n = 2$,

$$r_n = 0.529 \times \frac{2^2}{2}$$

$$= 2 \times 0.529$$

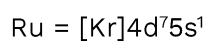
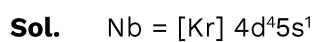
$$= 1.058 \text{ Å}$$

Q23 In Ru and Nb,



Then find the sum of x and y .

Ans. (11)



$$X = 4, y = 7$$

$$= 7 + 4 = 11$$